BTS –V(15) –11.17–1546 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

C

B.Tech. Degree V Semester Examination November 2017

SE 15-1505 MANUFACTURING PROCESSES
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) How is pig iron produced from iron ore?
(b) What do you mean by (i) hardness (ii) toughness?
(c) Explain the necessity of case hardening.
(d) What are the different flames used in gas welding?
(e) What are the merits of laser beam welding over the conventional welding techniques?
(f) What do you mean by Heat Affected Zone (HAZ)?
(g) Explain the following terms in sand casting with a neat sketch.
(i) runner (ii) riser (iii) sprue (iv) gate.
(h) Discuss how centrifugal casting is carried out.
(i) "Tool life will be more for orthogonal cutting than oblique cutting". Do you agree to this statement? Justify your answer.
(j) What do you mean by hot rolling and cold rolling? Which one will you choose if you want better surface finish? Why?

PART B

$(4 \times 10 = 40)$

II. (a) Draw an iron carbon equilibrium diagram and mark the zones of the following heat treatment processes. (6)
(i) Annealing (ii) Normalizing (iii) spherodising.
(b) Explain different alloys of copper and their applications. (4)

OR

III. (a) How do you differentiate between cast iron and steel based on the percentage carbon content? Discuss different types of cast iron. (5)
(b) What is tempering? What are the objectives of tempering operation? (5)

IV. (a) Explain Metal Inert Gas welding. How is it different from Tungsten Inert Gas welding? (5)
(b) Explain explosive welding with a neat sketch. Why is it called a solid state metal welding process? (5)

OR

V. (a) Explain the different weld defects. (5)
(b) Discuss how weld defects can be identified by NDT techniques. (5)

VI. (a) Explain slush casting with a neat sketch. Discuss its applications. (5)
(b) What is pressure die casting? What are the applications requiring pressure die casting? (5)

OR

VII. (a) What is pattern? Why are pattern dimensions not exactly the same of the product? (5)
(b) What are the desirable properties of moulding sand? (5)

VIII. (a) Explain different types of rolling mills. (5)
(b) What is extrusion? What do you mean by direct and indirect types of extrusions? (5)

OR

IX. (a) Explain different non-conventional machining operations in detail. (5)
(b) What are lathe accessories and attachments? Explain the lathe accessories in detail. (5)

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